

Part A: Authentication Log

Assignment 5

Exercise 1: To print

Write sed commands to:

- Print the first log entry.

```
sed -n '1p' auth.log
```

- Print the fifth log entry.

```
sed -n '5p' auth.log
```

- Print entries from lines 2 to 6.

```
sed -n '2,6p' auth.log
```

- Print the last entry.

```
sed -n '$p' auth.log
```

- Print every second log entry starting from line 1.

```
sed -n '1~2p' auth.log
```

- Print all entries containing Failed password.

```
sed -n '/Failed password/p' auth.log
```

- Print all successful login entries containing Accepted password.

```
sed -n '/Accepted password/p' auth.log
```

- Print entries from the first failed login to the authentication failure.

```
sed -n '/Failed password/,/authentication failure/p' auth.log
```

Exercise 2: Filtering Security Events

Write commands to:

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- Display failed login attempts.

```
sed -n '/Failed password/p' auth.log
```

- Display successful login attempts.

```
sed -n '/Accepted password/p' auth.log
```

- Display entries associated with the root account.

```
sed -n '/root/p' auth.log
```

- Display entries containing Invalid user.

```
sed -n '/Invalid user/p' auth.log
```

- Display all events related to IP address 172.16.0.8.

```
sed -n '/172.16.0.8/p' auth.log
```

- Display all events that do not contain Accepted password.

```
sed -n '/Accepted password/!p' auth.log
```

- Display all events except those generated by sudo.

```
sed -n '/sudo/!p' auth.log
```

- Display only the entries that contain either Failed password or Invalid user using multiple -e commands.

```
sed -n -e '/Failed password/p' -e '/Invalid user/p' auth.log
```

Exercise 3: Removing Irrelevant Entries

Without modifying the original file, display the log after:

- Deleting successful login entries.

```
sed '/Accepted password/d' auth.log
```

- Deleting entries containing Connection closed.

```
sed '/Connection closed/d' auth.log
```

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- Deleting the first three entries.

```
sed '1,3d' auth.log
```

- Deleting entries from lines 4 to 7.

```
sed '4,7d' auth.log
```

- Deleting all entries associated with 192.168.1.15.

```
sed '/192.168.1.15/d' auth.log
```

- Deleting every second entry starting from the first line.

```
sed '1~2d' auth.log
```

- Deleting all events that do not contain Failed password.

```
sed '/Failed password/!d' auth.log
```

Exercise 4: Classifying Security Events

- Replace Accepted password with [SUCCESSFUL LOGIN].

```
sed 's/Accepted password/[SUCCESSFUL LOGIN]/g' auth.log
```

- Replace Failed password with [FAILED LOGIN].

```
sed 's/Failed password/[FAILED LOGIN]/g' auth.log
```

- Replace Invalid user with [INVALID USER].

```
sed 's/Invalid user/[INVALID USER]/g' auth.log
```

- Replace authentication failure with [AUTHENTICATION FAILURE].

```
sed 's/authentication failure/[AUTHENTICATION FAILURE]/g' auth.log
```

- Replace root with PRIVILEGED_USER.

```
sed 's/root/PRIVILEGED_USER/g' auth.log
```

- Replace every occurrence of server with authentication-server.

```
sed 's/server/authentication-server/g' auth.log
```

Exercise 5: Masking Sensitive Information

Use sed to anonymise the log before sharing it.

- Replace IP address 10.0.0.25 with X.X.X.X.

```
sed 's/10\.0\.0\.25/X.X.X.X/g' auth.log
```

- Replace 172.16.0.8 with SUSPICIOUS_IP.

```
sed 's/172\.16\.0\.8/SUSPICIOUS_IP/g' auth.log
```

- Replace username admin with USER_A.

```
sed 's/admin/USER_A/g' auth.log
```

- Replace username student with USER_B.

```
sed 's/student/USER_B/g' auth.log
```

- Replace all digits with X.

```
sed 's/[0-9]/X/g' auth.log
```

- Replace periods in IP addresses with hyphens.

```
sed -E 's/([0-9]+\.[0-9]+\.[0-9]+\.[0-9]+)/\1-\2-\3-\4/g' auth.log
```

- Replace all IP addresses in the sample file with [REDACTED_IP].

```
sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log
```

Part B: Web-Server Log

Exercise 6:

Use access.log to solve the below questions

Write sed commands to:

- Print all requests that returned status code 200.

```
sed -n '/200/p' access.log
```

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- Print all requests that returned status code 401.

```
sed -n '/ 401/p' access.log
```

- Print all requests that returned status code 403.

```
sed -n '/ 403/p' access.log
```

- Print all requests to /admin/login.

```
sed -n '/\admin\login/p' access.log
```

- Print all POST requests.

```
sed -n '/POST/p' access.log
```

- Print requests containing ../.

```
sed -n '/\.\.\//p' access.log
```

- Print requests containing <script>.

```
sed -n '/<script>/p' access.log
```

- Print requests containing OR.

```
sed -n '/OR/p' access.log
```

- Print all requests except successful requests.

```
sed -n '/200/!p' access.log
```

- Print all requests from 10.0.0.25.

```
sed -n '/^10\.0\.0\.25/p' access.log
```

Exercise 7: Labelling Suspicious Web Requests

Use substitution to label the events:

- Replace status code 200 with [ALLOWED].

```
sed 's/200/ [ALLOWED]/' access.log
```

- Replace status code 401 with [UNAUTHORIZED].

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```
sed 's/401/ [UNAUTHORIZED]/' access.log
```

- Replace status code 403 with [BLOCKED].

```
sed 's/ 403/ [BLOCKED]/' access.log
```

- Replace /admin/login with /RESTRICTED_RESOURCE.

```
sed 's/\\admin\\login\\/\\RESTRICTED_RESOURCE/g' access.log
```

- Replace GET with READ_REQUEST.

```
sed 's/GET/READ_REQUEST/g' access.log
```

- Replace POST with SUBMIT_REQUEST.

```
sed 's/GET/READ_REQUEST/g' access.log
```

Exercise 8: Grouping Security Operations

- Use { } to perform multiple commands.

For entries containing status code 401:

Replace 401 with [UNAUTHORIZED].

Print the modified line.

```
sed -n '/ 401/ { s/401/[UNAUTHORIZED]/; p; }' access.log
```

- For entries containing status code 403:

Replace 403 with [BLOCKED].

Replace the source IP address with [REDACTED_IP].

Print the modified entry.

```
sed -n -E '/ 403/ { s/403/[BLOCKED]/; s/^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/; p; }' access.log
```

- For requests to /admin/login:

Replace the URL with /PROTECTED_PAGE.

Replace the source IP with [REDACTED_IP].

Print the result.

```
sed -n -E '/\//admin\//login/ { s/^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/; s/\//admin\//login\//PROTECTED_PAGE/g; p; }' access.log
```

Exercise 9: Adding Incident Labels

- Insert the following heading before the first line of auth.log:
SECURITY AUTHENTICATION LOG

```
sed '1i SECURITY AUTHENTICATION LOG' auth.log
```

- Append the following message after the final entry:
END OF SECURITY LOG

```
sed '$a END OF SECURITY LOG' auth.log
```

- Insert [WARNING] before every failed login.

```
sed '/Failed password/i [WARNING]' auth.log
```

- Append Review this event after every invalid-user entry.

```
sed '/Invalid user/a Review this event' auth.log
```

- Insert [SUSPICIOUS WEB REQUEST] before every 403 entry.

```
sed '/ 403/i [SUSPICIOUS WEB REQUEST]' access.log
```

- Append [REQUIRES INVESTIGATION] after every 401 entry.

```
sed '/ 401/a [REQUIRES INVESTIGATION]' access.log
```

- Replace all entries containing Connection closed with:
[CONNECTION TERMINATED]

```
sed '/Connection closed/c [CONNECTION TERMINATED]' auth.log
```

- Replace all successful web requests with:
[NORMAL WEB REQUEST]

```
sed '/200/c [NORMAL WEB REQUEST]' access.log
```

Exercise 10: Separating Log Entries with G

Use G to:

- Add a blank line after every authentication event.

```
sed 'G' auth.log
```

- Add a blank line after every failed login.

```
sed '/Failed password/G' auth.log
```

- Add a blank line after every blocked web request.

```
sed '/ 403/G' access.log
```

- Add a blank line after every second entry starting with the first entry.

```
sed '1~2G' auth.log
```

Exercise 11:

- Write failed logins to failed_logins.txt.

```
sed -n '/Failed password/p' auth.log > failed_logins.txt
```

- Write successful logins to successful_logins.txt.

```
sed -n '/Accepted password/p' auth.log > successful_logins.txt
```

- Write invalid-user events to invalid_users.txt.

```
sed -n '/Invalid user/p' auth.log > invalid_users.txt
```

- Write 401 web events to unauthorized_requests.txt.

```
sed -n '/ 401$/p' access.log > unauthorized_requests.txt
```

- Write 403 web events to blocked_requests.txt.

```
sed -n '/ 403$/p' access.log > blocked_requests.txt
```

- Write all events related to 172.16.0.8 to suspicious_ip.txt.

```
sed -n '/172.16.0.8/p' auth.log > suspicious_ip.txt
```

Exercise 12:

- Label successful and failed logins.

```
sed -e 's/Accepted password/[SUCCESSFUL LOGIN]/g' \-e 's/Failed password/[FAILED LOGIN]/g' auth.log
```

- Label 401 and 403 web requests.

```
sed -e 's/401/[UNAUTHORIZED]/g' \-e 's/403/[BLOCKED]/g' access.log
```

- Delete successful logins and print failed logins.

```
sed -n -e '/Accepted password/d' -e '/Failed password/p' auth.log
```

- Replace root with [PRIVILEGED_USER] and mask all IP addresses.

```
sed -E -e 's/root/[PRIVILEGED_USER]/g' \-e 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log
```

- Replace GET and POST with READ_REQUEST and SUBMIT_REQUEST.

```
sed -e 's/GET/READ_REQUEST/g' \-e 's/POST/SUBMIT_REQUEST/g' access.log
```

Exercise 13:

Create security_report.sed containing commands to:

Insert SECURITY INCIDENT REPORT before the first line.

Replace Failed password with [FAILED LOGIN].

Replace Invalid user with [INVALID USER ATTEMPT].

Replace Accepted password with [SUCCESSFUL LOGIN].

Replace all sample IP addresses with [REDACTED_IP].

Append END OF INCIDENT REPORT after the last line.

security_report.sed:

```
1i SECURITY INCIDENT REPORT
s/Failed password/[FAILED LOGIN]/g
s/Invalid user/[INVALID USER ATTEMPT]/g
s/Accepted password/[SUCCESSFUL LOGIN]/g
s/[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-
```

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```
9]\{1,3\}/[REDACTED_IP]/g
$a END OF INCIDENT REPORT
```

Execute it using:

```
sed -f security_report.sed auth.log
```

Save the output:

```
sed -f security_report.sed auth.log > incident_report.txt
```

Exercise 14: Modifying a Security Configuration File

- Create security.conf:

```
PermitRootLogin yes
PasswordAuthentication yes
MaxAuthTries 6
FirewallStatus disabled
LogLevel INFO
GuestLogin enabled
```

First make a backup:

```
cp security.conf security.conf.backup
```

- Use sed -i to:

Change PermitRootLogin yes to PermitRootLogin no.

```
sed -i 's/PermitRootLogin yes/PermitRootLogin no/' security.conf
```

Change MaxAuth Tries 6 to MaxAuthTries 3.

```
sed -i 's/MaxAuthTries 6/MaxAuthTries 3/' security.conf
```

Change FirewallStatus disabled to FirewallStatus enabled.

```
sed -i 's/FirewallStatus disabled/FirewallStatus enabled/'
security.conf
```

Change LogLevel INFO to LogLevel VERBOSE.

```
sed -i 's/LogLevel INFO/LogLevel VERBOSE/' security.conf
```

Change GuestLogin enabled to GuestLogin disabled.

```
sed -i 's/GuestLogin enabled/GuestLogin disabled/' security.conf
```

- Use the safer backup form:

```
sed -i.bak 'sed-command' security.conf
```

This exercise demonstrates configuration editing only; it does not configure the actual system.

Exercise 15:

- Write a Bash script named security_log_analyser.sh.
- The script should:

Accept a log filename as a command-line argument.

Check whether the argument is provided.

Check whether the file exists.

Extract failed login attempts.

Extract invalid-user attempts.

Replace sensitive IP addresses with [REDACTED_IP].

Add the heading SECURITY ALERT REPORT.

Add [ALERT] before suspicious entries.

Append END OF REPORT.

Save the result in security_alert_report.txt.

Display the generated report.

```
#!/usr/bin/bash
if [ $# -eq 0 ]
then
    echo "Error: Please provide a log filename."
    exit 1
fi
if [ ! -f "$1" ]
then
    echo "Error: File does not exist."
    exit 1
fi
echo "SECURITY ALERT REPORT" > security_alert_report.txt
sed -n \
```

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```
-e '/Failed password/p' \  
-e '/Invalid user/p' "$1" |  
sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' |  
sed 's/^[ALERT] /' >> security_alert_report.txt  
echo "END OF REPORT" >> security_alert_report.txt  
cat security_alert_report.txt
```